

Assignment 1 – Empirical Analysis of Fitts' Law

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- **Experiment Video Link:**

https://youtu.be/ioDor7_ZuRc

- **Trial Data Table:**

Fitts Law Experiment - Computer Science Field Guide

Trial	Average Distance (pixels)	Width (pixels)	Index of Difficulty (bits)	Average Time (milliseconds)
1	1006.09	92	3.58	771.91
2	529.7	277	1.54	448.6
3	266.4	92	1.96	478
4	599.4	185	2.08	533.3
5	888.5	185	2.54	544.4
6	321.3	185	1.45	368.7
7	329	277	1.13	345.3
8	631.1	92	2.97	577.1
9	755.1	277	1.9	436.3

- **Fitts' Law Formula & Analysis:**

The standard Shannon formulation of Fitts' Law is

$$MT = a + b \log_2 \left(\frac{A}{W} + 1 \right)$$

where (MT) represents movement time, (A) represents the distance to the target, and (W) represents the width of the target.

The Index of Difficulty is defined as: $ID = \log_2 \left(\frac{A}{W} + 1 \right)$

A linear regression was performed using Index of Difficulty as the independent variable and Movement Time as the dependent variable.

The resulting parameters were: $a = 164.79$ ms / $b = 157.73$ ms/bit

Therefore, the empirically derived Fitts' Law model for this experiment

is: $MT = 164.79 + 157.73 \log_2 \left(\frac{A}{W} + 1 \right)$

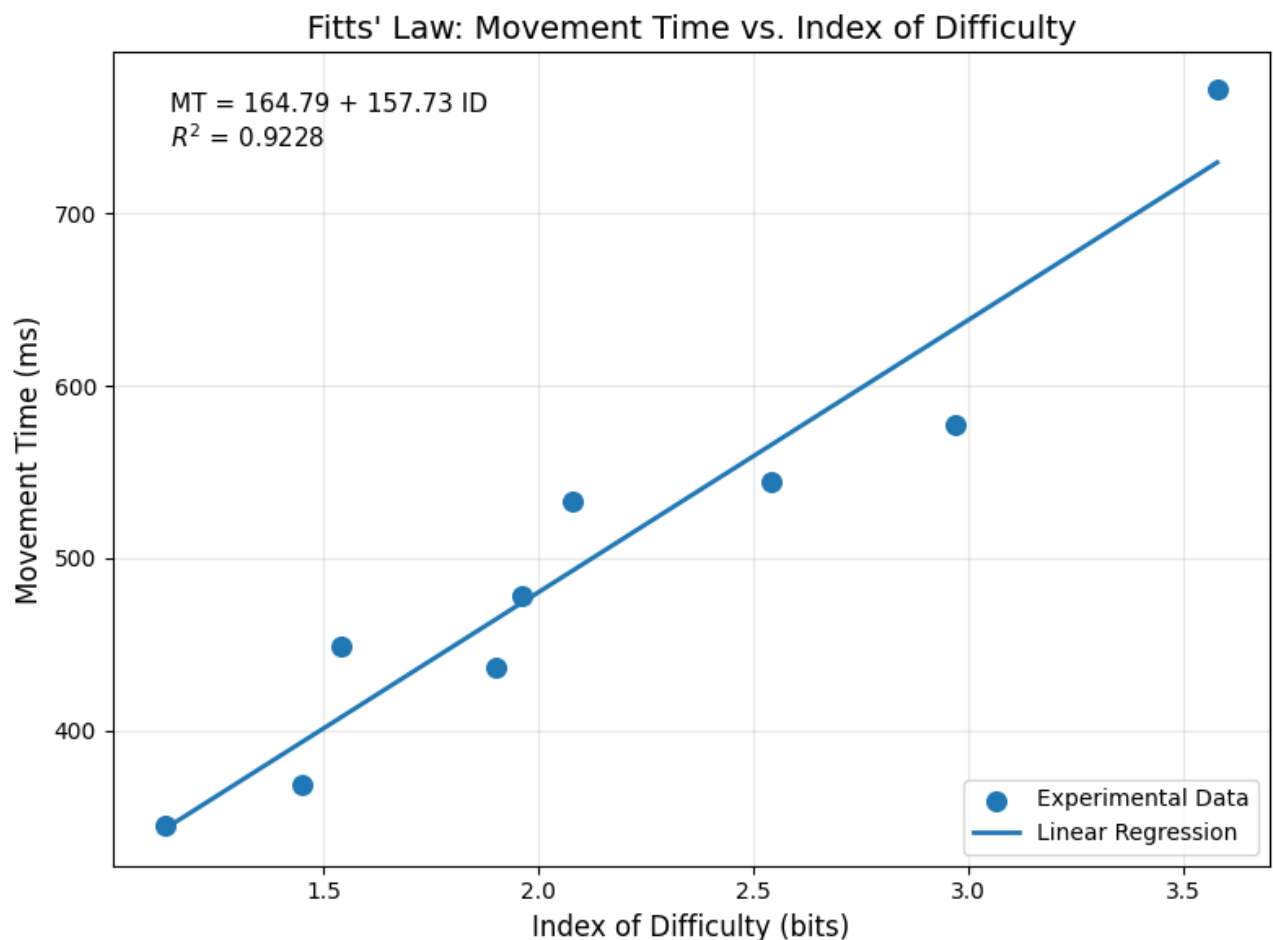
or equivalently: $MT = 164.79 + 157.73ID$

The intercept ($a=164.79$) ms represents the estimated baseline time required for basic perception, reaction, and motor execution when the Index of Difficulty approaches zero. The slope ($b=157.73$) ms/bit

indicates that for every one-bit increase in task difficulty, the predicted movement time increases by approximately 157.73 ms.

These parameters may vary among different users because of factors such as pointing-device familiarity, hand-eye coordination, motor ability, fatigue, input-device characteristics, display size, and other environmental or individual differences.

- **Scatter Plot:**



- **Conclusion:**

The experimental results show a strong linear relationship between Index of Difficulty and Movement Time. The obtained R^2 value of 0.9228 indicates that approximately 92.28% of the variation in Movement Time can be explained by the Fitts' Law model. Overall, the results support the prediction of Fitts' Law that pointing tasks require more time as the target becomes more difficult to acquire.